Data Mining and Machine Learning: Submission Details

F20DL/F21DL (2026/2027)

This document contains an in-depth breakdown of the deliverables you must submit, to help you understand the considerations and criteria outlined in the marking rubric.

Note: For your convenience, we've underlined the parts of this coursework descriptor that are applicable for F21DL only (i.e., PG students) ☺

The Report (D2)

Your report must be a **maximum of 6 pages** (not including title page, references, and appendices), and a **minimum font size of 11pts.** All reports must use **A4 paper format (21 cm x 29.7 cm).** Reports must not be submitted with any other paper size. All page margins (top, bottom, left, and right) must **not be smaller than 1.27 cm.**

Citations within the report can use any format: APA, Harvard, Numeric, IEEE, etc. However, regardless of choice used, **it must be consistent** throughout the report.

Place figures and tables in the paper near where they are first discussed. **Only include graphs that directly support your analysis**—avoid decorative or redundant visuals.[1] All visuals should be **clearly labelled, consistently formatted, and sized appropriately** without sacrificing legibility. **Number each figure and table sequentially** (e.g., "Figure 1: Caption", or "Table 1: Caption") and provide clear and descriptive captions. Each visual must be correctly **cross-referenced in the text** (e.g., "As shown in Table 1…").

0. Title Page

Your title page must contain the following information and **no other pertinent details.**

1. The title of your project
2. The name of your group
3. A link to your GitHub repository
4. The full name for each group member and their Heriot-Watt matriculation number (starting with "H")

You may include images and other design styles if you wish to.

1. Introduction

This section should be a **short paragraph that motivates the project**; for example, consider why it is useful and who would it be useful for.

We recommend that you connect your motivation with **a very brief overview of past research** and continue with an explanation of the problem(s) you are trying to solve through the various aspects of this coursework.

Importantly, **this section is not a deep exploration into the background** behind any specific methodology or prior work, nor is it an explanation as to why your research is tackling a completely novel question or reinventing the proverbial wheel.

[1] Refer to From Data To Viz for advice on different types of plots and common caveats that you should avoid doing.

By the end of this section, you should have **demonstrated a deep understanding of the subject matter**. The reader should have a clear understanding of your chosen field of research and know both why your solutions are needed and are a valuable contribution to the field.

PG Only

This section should *also* **introduce and justify your research questions.** Therefore, when stating your hypotheses, you should also **justify your expectations using prior literature.**

2. Related work (PG Only)

This section should delve deeper into the **relevant background surrounding your research question(s).** As a result, this section should **use existing literature—with citations—to justify your chosen approach** for each experiment. For example, how has prior work influenced your choice to use this classifier with this dataset? What kind of research gap are you addressing? What have others in this field done to address the same question(s)?

By the end of this section, the reader should have a clear understanding of why each of your solutions is apt given your research question(s) and the context of the wider research previously conducted on the topic.

If required, **you may use up to one additional page for your related work section only.**

3. Dataset Description and Analysis

This section should **explain and analyse your chosen dataset(s)**. You should start with a general description of the dataset(s) including details such as its source, the license(s) associated, the features provided (e.g., metadata or column names) and any relevant summary statistics to help communicate broad details across the entire dataset. Where possible and relevant, use citation to source your dataset(s).

Finally, state any peculiarities that need to be addressed. For example, is the data skewed or unbalanced in any way? From these details, describe and justify the data cleaning and preprocessing steps you took, such as removing or transforming features. You can choose to use tables and/or graphs to describe/illustrate any of the above points.

By the end of this section, the reader should have a clear understanding of both how and why your chosen dataset(s) will help answer your chosen research question(s), how they are structured, and all steps that were taken to preprocess the data, paired with reasonable explanations as to why each step was needed.

4. Experimental Setup

This section should describe your choice of algorithmic solutions.

Having explained your research question(s) in the previous sections, you should now explain how each algorithm was used. For example, what was your architectural choice for tackling the classifier component of the assignment? How did you arrive at choosing the number of clusters for your K-Means algorithm (if you used a K-means algorithm)? Which metrics were used to evaluate performance of the neural networks? Were there any particularities with preliminary experiments that necessitated further tweaks (e.g., either architectural, or hyperparameter related?)

By the end of the section, the reader should have a clear understanding of which solutions you employed to deal with each experimental question, alongside clearly justified explanations as to training related nuances for each.

5. Results

This section presents the outcome from all experiments you have run. For each experiment, provide:

- A clear textual description of the results and the found effects
- Comparisons between different experimental outcomes (e.g., “Contrary to the classifier, the neural network showed...”)
- Direct connections to your research question(s)
- Explicit statements whether or not your original hypotheses/assumptions are supported by your findings

Additionally, state any further analyses/experiments/ablations that you ran which were not stated in the above sections, alongside clear justifications as to why each was needed (e.g., “To understand the low overall performance, we conducted an exploration of F1 scores by each feature and found that X and Y were challenging for the model, as seen in the breakdown of recall/precision in Figure K”).

Each experiment should be explained clearly through tables and/or graphs that are correctly labelled, referenced, and formatted in a consistent manner.

By the end of this section, the reader should have a comprehensive understanding of all experimental outcomes, **supported by both textual explanations and visual evidence.** Importantly, **this section should focus on presenting results**. Save interpretations, discussions of why certain approaches succeeded or fails, and broader implications for the Discussion section.

6. Discussion

This section should explain the findings and their implications in relation to your research question(s) and objectives.

Start by explaining your findings. while linking the outcomes between your results and your original research aims. For example, are your classifier’s results in line with your expectations (if you had any)? Why/Why not? Do prior data analyses help with explaining these results?

Lastly, **discuss the practical implications of your work.** How realistic is your proposed solution? How reliable and robust would your models be if deployed? If there are risks or weaknesses that may arise, how could these be mitigated with further development, if at all? Alternatively, if there are no risks, outline the steps taken to ensure your approach is safe for deployment?

By the end of this section, the reader should have a clear understanding as to why your experiments led to your results. Remember to **maintain a balanced perspective,** acknowledging both the strengths and limitations of your work while positioning it within the broader research context.

PG Only

Additionally, this section should also explain the implications of your findings within the wider research sphere. When explaining your findings, you **must also link your results and research question(s) to the existing literature.** For example, what can you find in the literature that might explain the various effects found? Where possible, **consider framing your findings** within the context of existing knowledge, **drawing parallels or noting any departures from existing work**.

In this section, also **address the limitations of your work** to help guide future research. For example, was the data appropriate for your clustering approach? What further steps should future researchers take to continue your work? For example, “Future research should gather X type of data to address X limitation we encountered, as seen in...” Importantly, you should provide **specific, actionable recommendations that build upon your findings** and address the gaps you identify.

By the end of this section, along with a clear understanding of your experiments, readers will also have a clear direction for how future work would move beyond your implementation and overcome challenges that you faced. Importantly, you **must support your arguments with relevant citations** to strengthen your claims and justifications.

7. Conclusion

This section should **briefly summarise your study.** By reading this section, a reader should understand **why you were motivated** to do your research, **why you chose your datasets**, what algorithmic choices you made, your results and how they relate with your research question(s), and finally what **the take-aways of the discussion section** were.

Avoid highly technical details, extensive citations, or referring to specific numerical results.

8. References

Gather the full set of references together and place in this section. Place the References section before any Appendices. You may use a bibliography tool to manage and insert citations, but this is not mandatory. The format of your references **must be compatible with your chosen citation style**—for example, do not use a numerical citation style with APA references. As with the citations, regardless of your choice, it must be consistently applied.

9. Appendices

It may be the case that aspects of your report are exhaustive. If this is the case, you should consider moving less pertinent details to the appendix. However, **your report must be self-contained: it is optional for readers to look at the appendices.** Content included within the appendix **must not be critical** to the reading and understanding of the report.

While there is no length limitation to the appendices, **you should not include unnecessary details, code, or any content that is not pertinent to the content of the report.**

Where necessary, you should refer to the specific sections within the Appendix. For example, you should do the following:

> Refer to Appendix A.2 for implementation details.

The Presentation (D3)

Alongside your report, you must also give a presentation. As mentioned in the coursework description, each group **only has 15 minutes** to present and discuss their findings. You will have a **maximum of 7 minutes** to present a short slide deck, with a **maximum of 5 slides**, followed by an **8-minute Q&A session** conducted by members of staff/lab helpers.

We strongly encourage all group members to speak at least once during the 15-minute presentation (either during the slides and/or the Q&A session) to demonstrate that all group members share a clear understanding of the work done.

We highly recommend that each group is prepared to present, with your presentation, before the presentation starts. You can present using a laptop/tablet screen (provided it's not too small) as you will not be standing far from us, and we can lean in to read more carefully.

Additionally, prior to your presentation, we recommend knowing who you are presenting to so that you can **wait nearby 5 minutes prior to your start time.**

You do not need to submit your presentation or include it within the repository. **This means that you are allowed to work on your presentation after the report submission.**